Model Validation, Overfitting Control & Hyperparameter Tuning

Maincrafts Technology AIML Internship - Task 3

Introduction

This report summarizes a professional validation workflow for the California Housing prediction problem. The project extends the earlier model-comparison workflow with overfitting analysis, cross-validation, GridSearchCV tuning, optimized model evaluation, and reproducible model persistence.

Objective

The objective is to detect overfitting, estimate model generalization with five-fold cross-validation, tune Decision Tree hyperparameters, and select the final regression model using RMSE and R2.

Methodology

The dataset contains 20,640 rows, 8 features, and the target variable HousePrice. Features were scaled with StandardScaler and split into 16,512 training rows and 4,128 test rows using random_state=42. Baselines include Linear Regression, Ridge Regression, and the Task 2-style max_depth=5 Decision Tree.

Overfitting Analysis

The original Decision Tree achieved train RMSE 0.6959 and test RMSE 0.7242, a gap of 0.0283. The lower training error indicates some overfitting, while the controlled max_depth keeps the gap from becoming extreme.

Cross Validation

Five-fold cross-validation produced a mean RMSE of 0.8141 with standard deviation 0.0563. This is more reliable than a single split because every fold becomes validation data once and the result reflects variability across multiple partitions.

GridSearchCV

GridSearchCV tuned max_depth values [3, 5, 7, 10] and min_samples_split values [2, 5, 10] with cv=5. The best parameters were {'max_depth': 10, 'min_samples_split': 10} with best CV RMSE 0.6366.

Optimized Model

The best final model was Tuned Decision Tree with test RMSE 0.6454 and R2 0.6821. It was selected by the rule: highest R2 Score, then lowest RMSE.

Conclusion

The tuned Decision Tree balances predictive performance and generalization by controlling tree complexity with cross-validated hyperparameters. The workflow generates reusable artifacts, transparent metrics, and validation evidence suitable for portfolio review.

Results Summary

Model ranking

Rank	Model	RMSE	R2
1	Tuned Decision Tree	0.6454	0.6821
2	Original Decision Tree	0.7242	0.5997
3	Ridge Regression	0.7456	0.5758
4	Linear Regression	0.7456	0.5758

Cross-validation fold RMSE

Fold 1: 0.8509

Fold 2: 0.7457

Fold 3: 0.7558

Fold 4: 0.8935

Fold 5: 0.8244

Mean: 0.8141

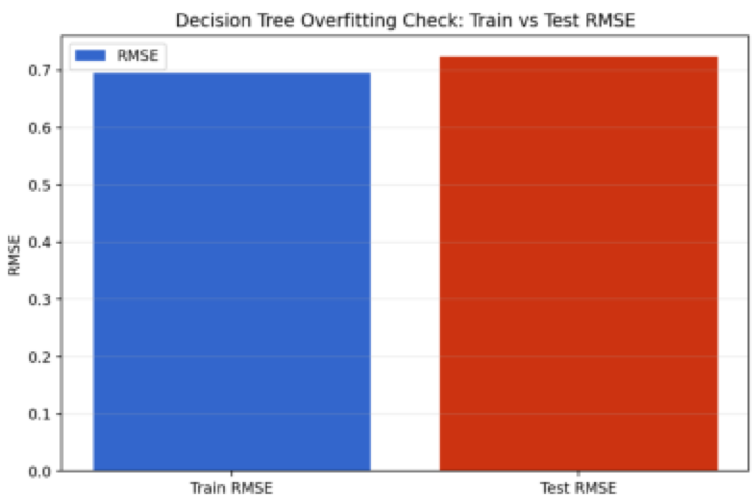
Std: 0.0563

Model Justification

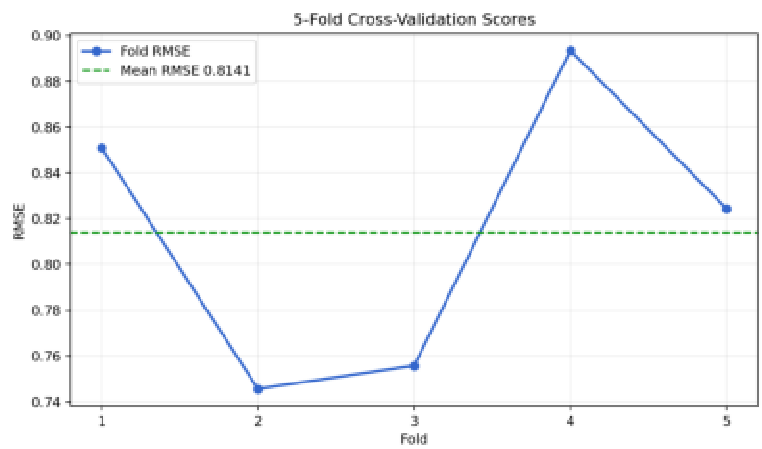
Model justification: The final model was selected because it delivered the strongest held-out test performance after complexity controls were tuned through cross-validation. GridSearchCV improves confidence by avoiding manual hyperparameter choice and comparing candidate settings on multiple validation folds. The chosen depth and split threshold reduce variance compared with a deeper unrestricted tree while preserving non-linear predictive power.

Task 3 Visualizations

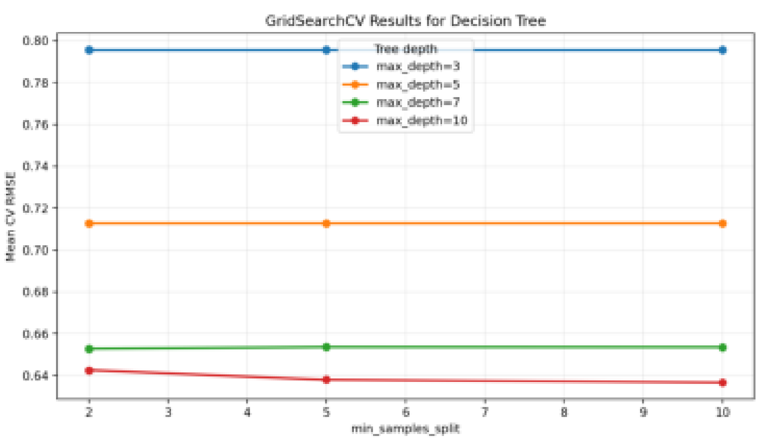
Train vs Test RMSE



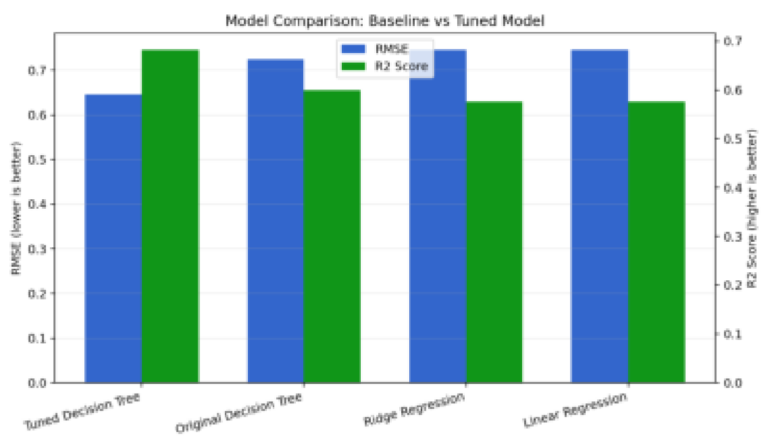
Cross-Validation Scores



Grid Search Results



Model Comparison



Feature Importance